

Solutions

1. **Solution:** Domain $\{-2, 0, 4\}$; Range $\{3, -1, 7\}$.

2. **Solution:** Domain $\{-2, -1, 0, 1, 2, 3\}$; Range $\{-3, -1, 1, 3, 5, 7\}$ (plug into $y = 2x + 1$).

3. **Solution:** Domain $[-1, 4]$; Range $[-2, \infty)$.

4. **Solution:** $m = \frac{6 - (-2)}{7 - 3} = \frac{8}{4} = 2$ (rising line).

5. **Solution:** $m = \frac{\Delta y}{\Delta x} = \frac{-10}{4} = -\frac{5}{2}$ (negative).

6. **Solution:** (a) slope 0, equation $y = -3$. (b) slope undefined, equation $x = 4$.

7. **Solution:** Use point-slope: $y + 5 = 3(x - 2) \Rightarrow y = 3x - 11$.

8. **Solution:** $4 = (-\frac{2}{3})(6) + b = -4 + b \Rightarrow b = 8$. Equation $y = -\frac{2}{3}x + 8$.

9. **Solution:** $m = \frac{-4 - 2}{5 - (-1)} = \frac{-6}{6} = -1$; then $y = -x + b$ and $2 = 1 + b \Rightarrow b = 1$. So $y = -x + 1$.

10. **Solution:** Point-slope: $y - 7 = -\frac{1}{2}(x + 4)$. Slope-intercept: $y = -\frac{1}{2}x + 5$.

11. **Solution:** $m = \frac{11 - (-1)}{9 - 3} = \frac{12}{6} = 2$. Point-slope: $y + 1 = 2(x - 3)$. Slope-intercept: $y = 2x - 7$.

12. **Solution:** At $x = 2$, $y = -3(2) + 4 = -2$. Point-slope: $y - (-2) = -3(x - 2)$, i.e., $y + 2 = -3(x - 2)$.

13. **Solution:** Differences: $7 - 4 = 3$, $10 - 7 = 3$, $14 - 10 = 4$ (not constant) $\Rightarrow$ not linear. Domain $\{0, 1, 2, 3\}$, Range $\{4, 7, 10, 14\}$.

14. **Solution:** $\Delta y = m \Delta x = \frac{2}{3} \cdot 9 = 6$. New $y = -3 + 6 = 3$.

15. **Solution:** With $b = -2$ and point $(4, 0)$, slope $m = \frac{0 - (-2)}{4 - 0} = \frac{2}{4} = \frac{1}{2}$. Same as L_1 with same $b \Rightarrow$ same line.

16. **Solution:** Slope $m = -\frac{3}{2}$; equation $y = -\frac{3}{2}x + 5$. For $y = -7$: $-7 = -\frac{3}{2}x + 5 \Rightarrow -12 = -\frac{3}{2}x \Rightarrow x = 8$.

17. **Solution:** $10 - 4 = m(-2 - 1) \Rightarrow 6 = -3m \Rightarrow m = -2$. Then $y - 4 = -2(x - 1) \Rightarrow y = -2x + 6$.

18. **Solution:** For $-2 \leq x \leq 3$, $y = 2x + 1$ runs from -3 to 7 ; plus the point $(5, 7)$ does not change the max. Domain $[-2, 3] \cup \{5\}$; Range $[-3, 7]$.

19. **Solution:** $m_{AB} = \frac{-2 - 4}{2 - (-1)} = -2$, $m_{BC} = \frac{-8 - (-2)}{5 - 2} = -2 \Rightarrow$ collinear. Use A : $y - 4 = -2(x + 1) \Rightarrow y = -2x + 2$.

20. **Solution:** Rates: $(-2 \rightarrow 0)$: $\frac{5 - 1}{0 - (-2)} = 2$; $(0 \rightarrow 3)$: $\frac{14 - 5}{3 - 0} = 3$; $(3 \rightarrow 7)$: $\frac{26 - 14}{7 - 3} = 3$. Not linear overall (rates differ). Average rate $(-2 \rightarrow 7)$: $\frac{26 - 1}{7 - (-2)} = \frac{25}{9}$.

21. **Solution:** Set $0 = mx + b \Rightarrow x = -\frac{b}{m} = -\frac{6}{-3/4} = 8$.

22. **Solution:** Points $(4, 0)$ and $(0, -6)$. $m = \frac{0 - (-6)}{4 - 0} = \frac{6}{4} = \frac{3}{2}$. Equation $y = \frac{3}{2}x - 6$.

23. **Solution:** Slope $m = \frac{(b + 3) - b}{(a + 1) - a} = 3$. Point-slope: $y - b = 3(x - a)$. Slope-intercept: $y = 3x + (b - 3a)$.